

A7 · One Hop at a Time

CS425 Computer Networks · instructor led · rounds 1 and 2 run on Onyx (ssh onyx), rounds 3 and 4 are paper

TURN THIS IN ON PAPER

before you leave the room
graded pass/fail

Names _____ Date _____

Every round: **predict** first, in ink, then **run** the command or work it on paper, then **check**. A wrong prediction costs nothing. A prediction written afterwards is not a prediction.

ROUND 1 · ONYX'S OTHER ADDRESS `ip -br link`

QUESTION	ANSWER
Predict: Onyx moves to another building and subnet. Which address changes, the IP or the MAC?	
Onyx's MAC address, from the output · its OUI (the first 3 bytes)	
First byte of Onyx's MAC in binary · unicast or multicast? global or local?	
First byte of a phone's randomized Wi-Fi address in binary · global or local?	

Q1. Why would a phone use a made up, locally administered address on each Wi-Fi network instead of the one burned into its hardware?

ROUND 2 · ARP `ip neigh` · `ping -c 1 8.8.8.8`

QUESTION	ANSWER
Predict: after the ping, is <code>8.8.8.8</code> in the ARP table? If not, whose MAC goes in the frame?	
The gateway's MAC address, from <code>ip neigh show 132.178.227.1</code>	

THE FRAME LEAVING ONYX	ADDRESS	CHANGES AT THE NEXT HOP?
Source MAC		
Destination MAC		
Source IP		
Destination IP		

Q2. Why does Onyx never send an ARP request for `8.8.8.8`?

ROUND 3 · CRC BY HAND $D = 11010011$ · $G = 1011$

QUESTION	ANSWER
Predict: how many bits long is R?	
R · the 11 bits the sender puts on the wire	
Receiver's remainder for D then R · for the same 11 bits with the third bit flipped	

Divide 11010011000 by 1011 . Subtraction is XOR: no borrows, no carries.

ROUND 4 · BE THE SWITCH A, B, C, D on ports 1, 2, 3, 4

QUESTION	ANSWER
Predict: how many of the five frames get flooded?	

#	FRAME	OUT WHICH PORTS?	SWITCH TABLE AFTER (HOST: PORT)
1	A to D		
2	D to A		
3	B to D		
4	C to ff:ff:ff:ff:ff:ff		
5	A to C		

Q3. B runs Wireshark the whole time. Which frames does B see? Which would B see if a hub replaced the switch?

Exit. Loading `boisestate.edu` on a fresh laptop runs DHCP, ARP, DNS, TCP and HTTP, in that order. Which one never crosses a router, and why can it not?

Graded pass/fail on something in every box. Hand this in before you leave.